SYSTEM DESIGN

Caching & CDNs Visual Guide

Serve it once, serve it fast — everywhere.
How caches cut latency and cost.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Problem: Every Request Hits Origin

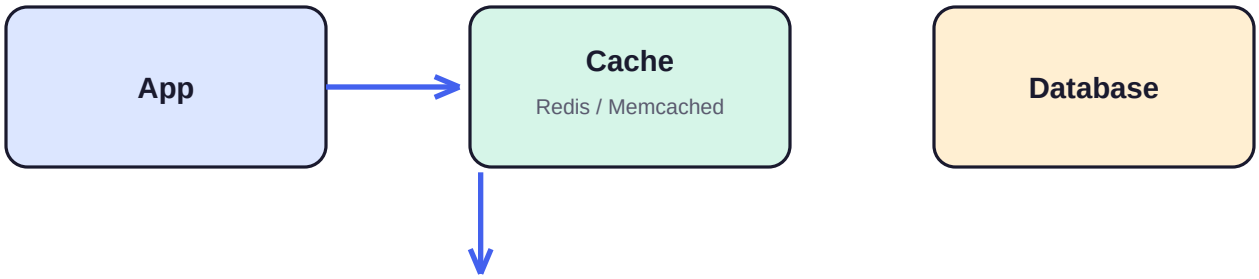
Without caching, each request travels the full path and recomputes everything.

Origin does redundant work per request. Latency = distance + compute, every single time.

STEP 2/6

Cache-Aside: Check the Cache First

The classic pattern — look it up before you compute it.



1. App asks cache for key
2a. HIT → return instantly
2b. MISS → query database
3. Store result in cache (TTL)
4. Return to caller

STEP 3/6

Cache Invalidation Strategies

The two hard problems: naming things, cache invalidation, off-by-one errors.

TTL Expiry

Entries auto-expire after N seconds. Simple; serves stale briefly.

Write-Through

Write to cache AND db together. Always fresh; slower writes.

Write-Behind

Write to cache now, db later. Fast; risk of loss.

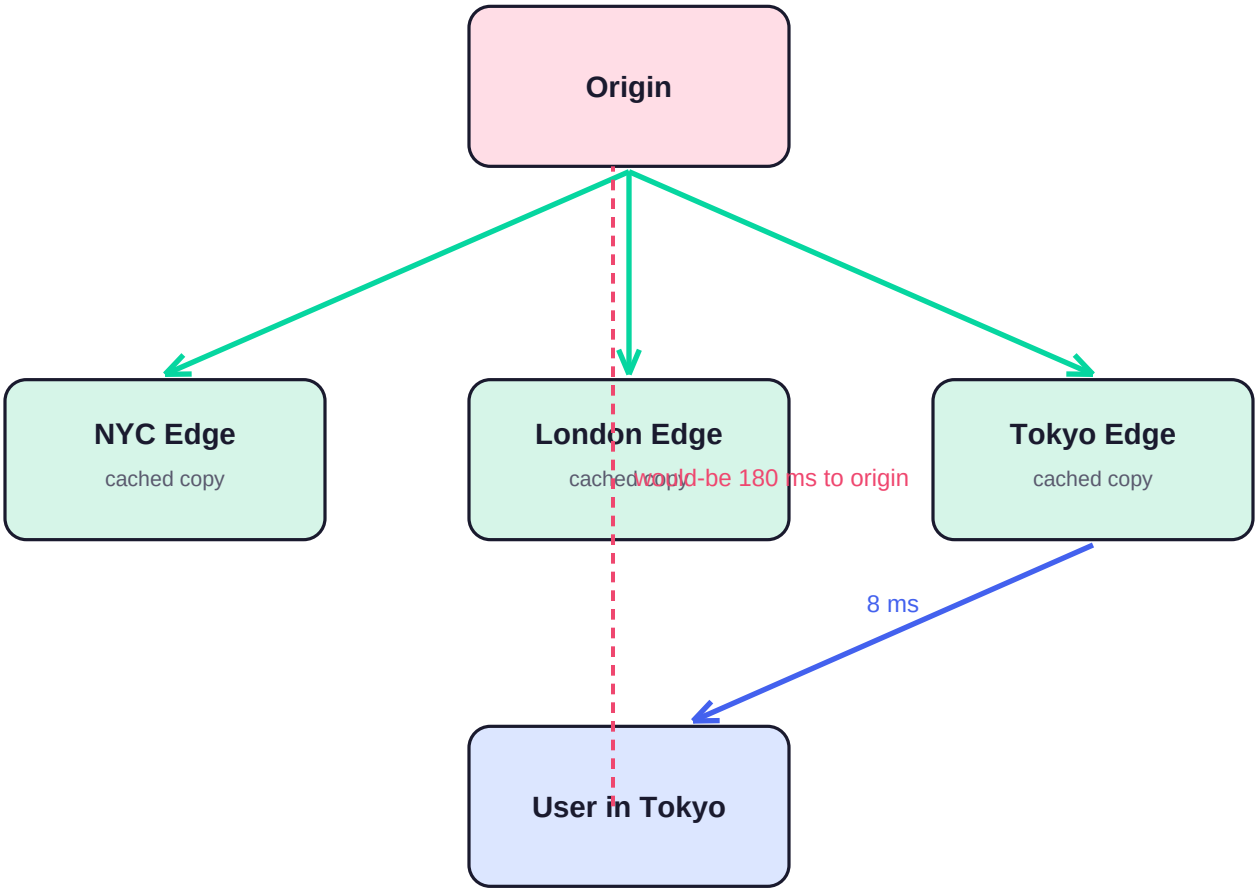
Eviction (LRU/LFU)

Full cache? Drop least-recently / least-frequently used.

STEP 4/6

CDN: Cache at the Edge

Push static content to servers near every user on Earth.



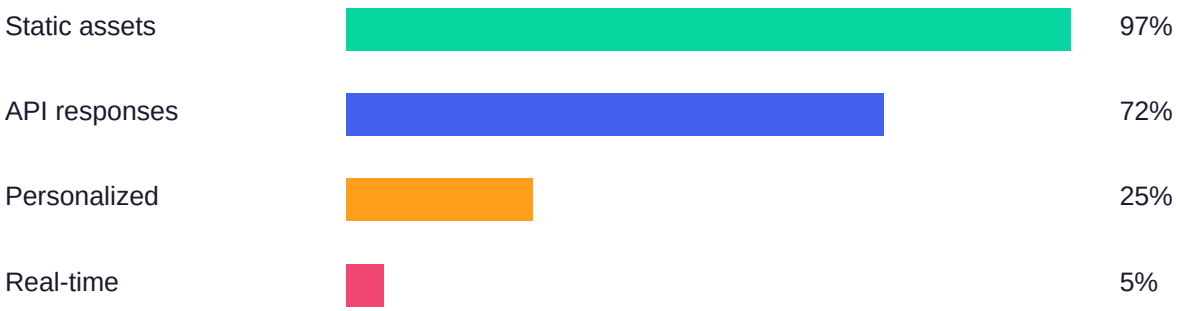
Edge caches serve 90%+ of static traffic locally. Origin only sees cache misses.

STEP 5/6

Cache Hit Ratio: The Scoreboard

Hit ratio = hits / (hits + misses). Higher is better.

Hit ratio by content type

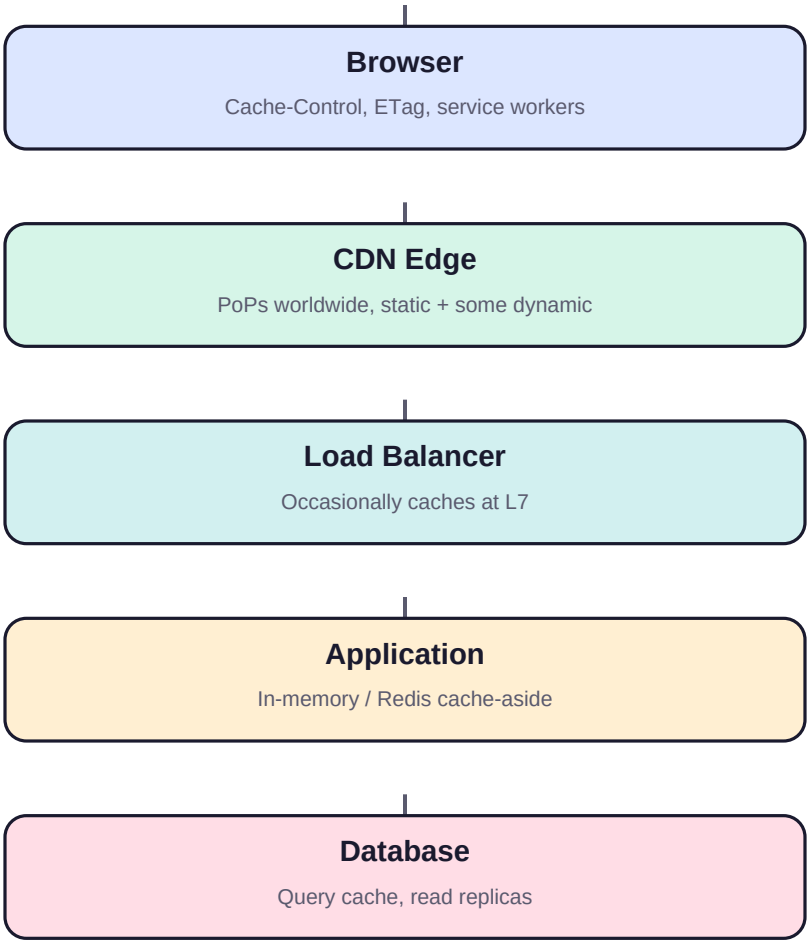


Static assets cache beautifully. Personalized and real-time data barely cache at all — design accordingly.

STEP 6/6

Full-Stack Caching Layers

Cache at every layer of the stack.



STEP ✓I✓

Key Takeaways

Caching & CDNs — Remember

- Cache-aside: check cache first; on miss, load from DB and populate with a TTL.
- Invalidation choices: TTL (simple, briefly stale), write-through (fresh, slower), write-behind (fast, risky).
- CDNs push content to edge locations — users fetch from nearby, not from origin.
- Hit ratio tells you if caching is working; static assets hit 95%+, personalized data far less.
- Layer your caches: browser → CDN → app → database.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev